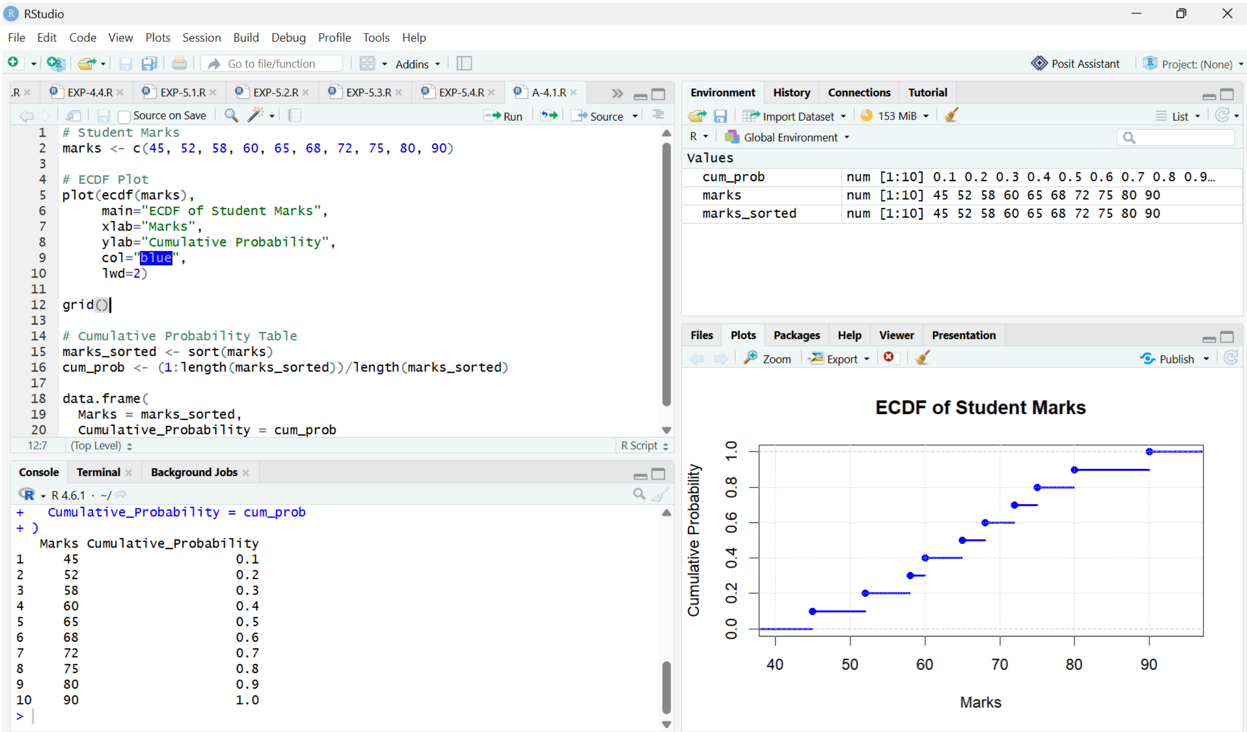
QUESTION-1:



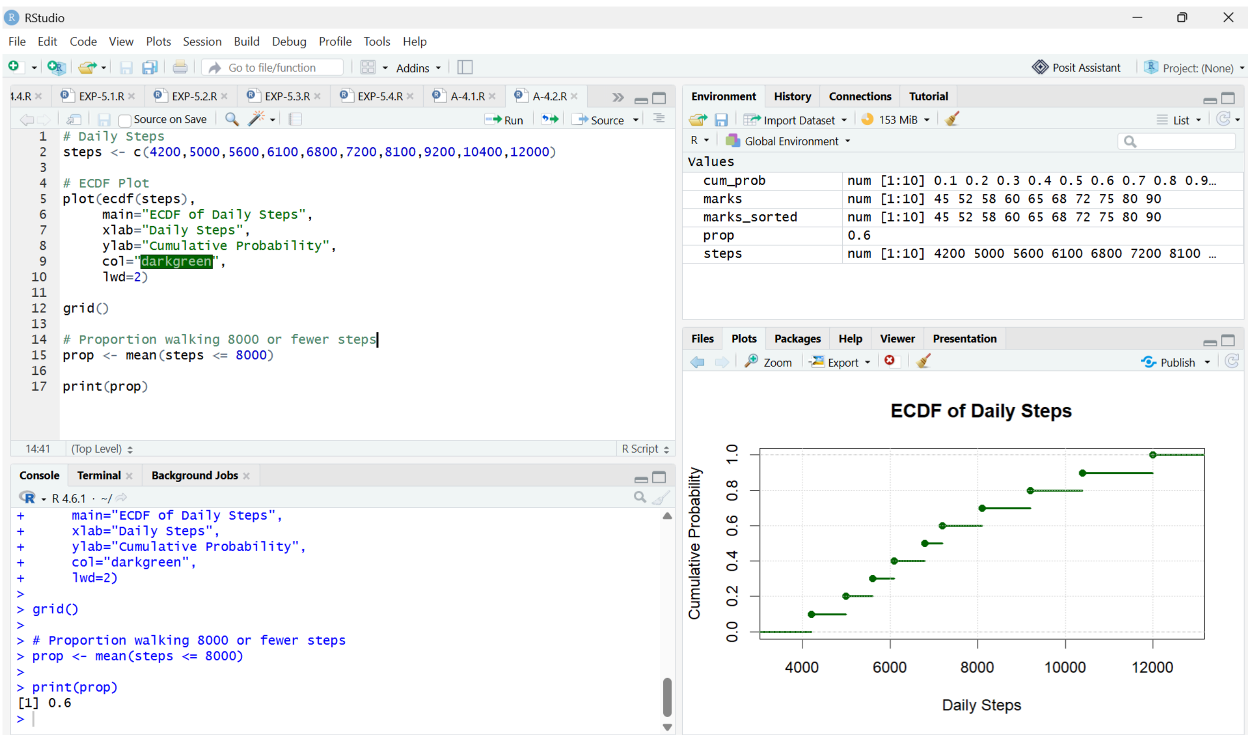
CODE:

```
# Student Marks
marks <- c(45, 52, 58, 60, 65, 68, 72, 75, 80, 90)

# ECDF Plot
plot(ecdf(marks),
     main="ECDF of Student Marks",
     xlab="Marks",
     ylab="Cumulative Probability",
     col="blue",
     lwd=2)

grid()

# Cumulative Probability Table
marks_sorted <- sort(marks)
cum_prob <- (1:length(marks_sorted))/length(marks_sorted)

data.frame(
  Marks = marks_sorted,
  Cumulative_Probability = cum_prob
```

Interpretation

- The ECDF increases step-by-step as marks increase.
- 50% of students scored **65 marks or less**.
- 80% of students scored **75 marks or less**.
- All students scored **90 marks or less**.

QUESTION-2



CODE:

Daily Steps

steps <- c(4200,5000,5600,6100,6800,7200,8100,9200,10400,12000)

ECDF Plot

plot(ecdf(steps),

main="ECDF of Daily Steps",

xlab="Daily Steps",

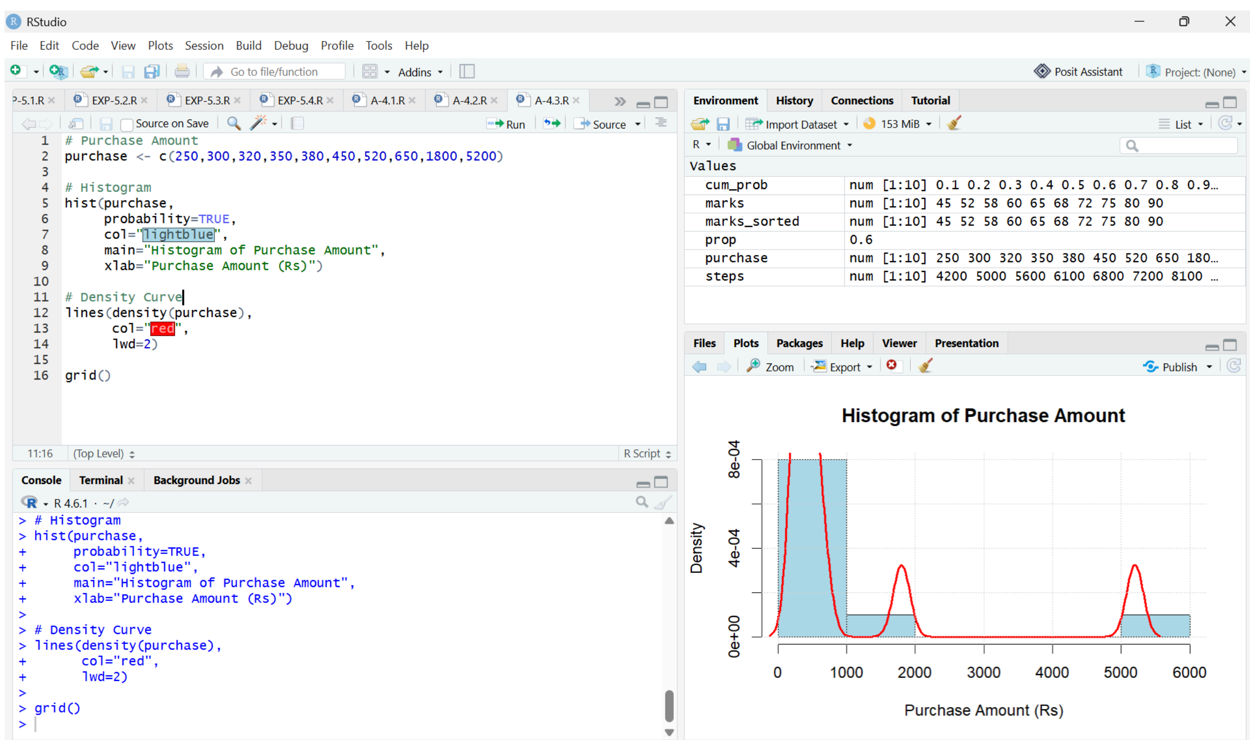
ylab="Cumulative Probability",

```
    col="darkgreen",
    lwd=2)
grid()
# Proportion walking 8000 or fewer steps
prop <- mean(steps <= 8000)
print(prop)
```

Interpretation

- **60%** of participants walked **8000 steps or fewer**.
- The ECDF helps determine the percentage of observations below any specified value.
- It is useful for comparing distributions and identifying percentiles.

QUESTION-3:



CODE:

```
# Purchase Amount
purchase <- c(250,300,320,350,380,450,520,650,1800,5200)
```

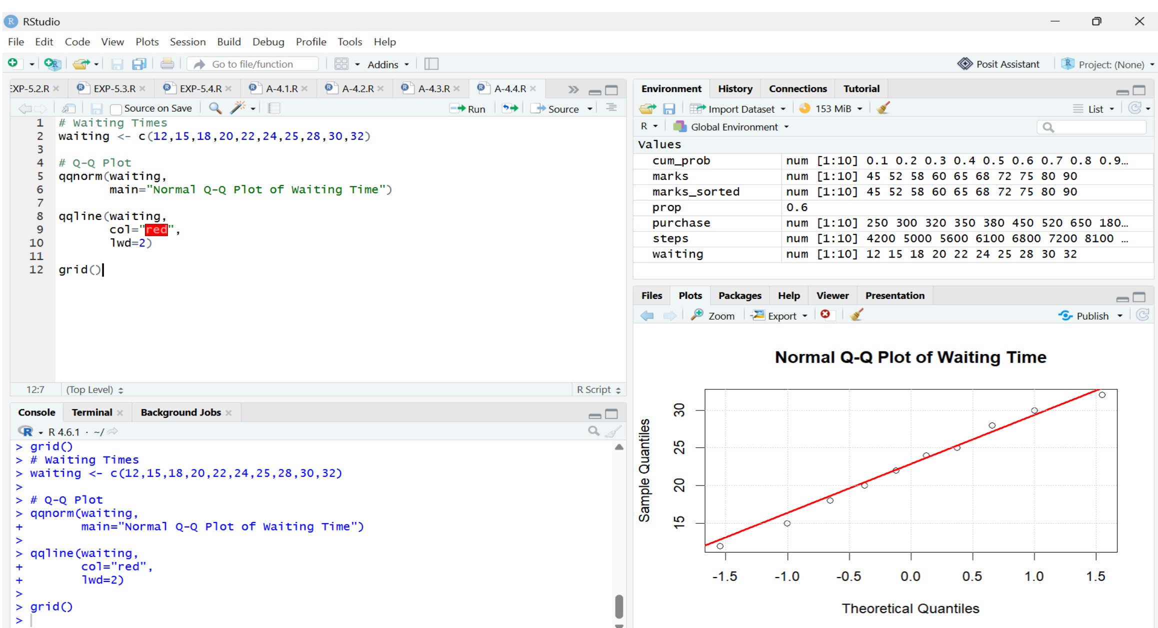
```
# Histogram
hist(purchase,
     probability=TRUE,
     col="lightblue",
     main="Histogram of Purchase Amount",
     xlab="Purchase Amount (Rs)")

# Density Curve
lines(density(purchase),
      col="red",
      lwd=2)

grid()
```

Interpretation

The distribution is **positively skewed (right-skewed)** because:

- Most purchase amounts are between **₹250 and ₹650**.
- Two large values (**₹1800 and ₹5200**) create a long right tail.
- The mean is greater than the median due to these high-value purchases.

QUESTION-4:



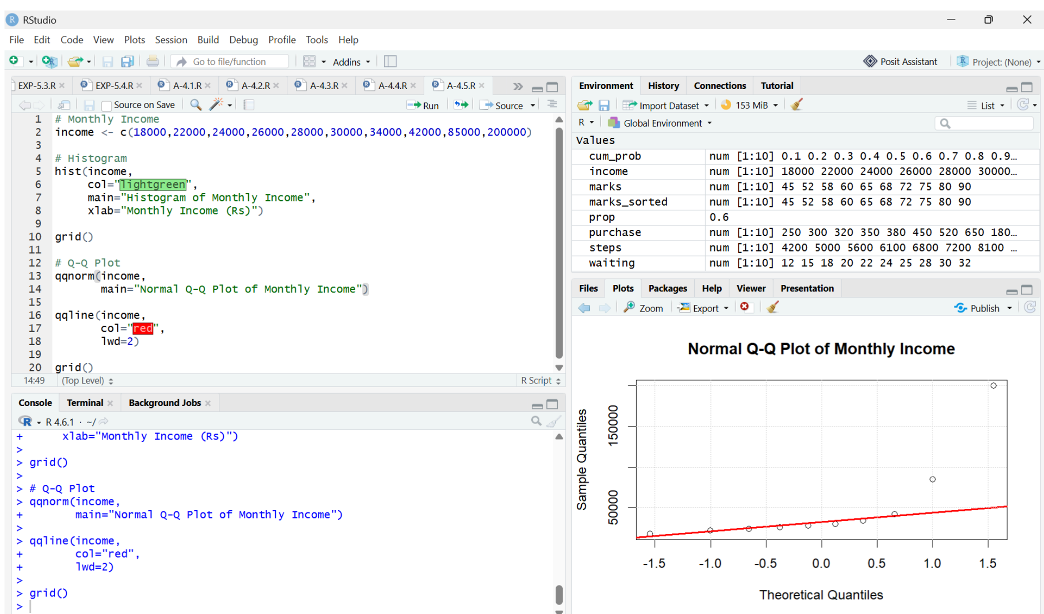
CODE:

```
# Waiting Times

waiting <- c(12,15,18,20,22,24,25,28,30,32)

# Q-Q Plot

qqnorm(waiting,

      main="Normal Q-Q Plot of Waiting Time")

qqline(waiting,

      col="red",

      lwd=2)

grid()
```

Interpretation:

- Most points lie close to the reference line.
- There are no major deviations.
- Therefore, the waiting time data is **approximately normally distributed**.

QUESTION-5:



CODE:

```
# Monthly Income

income <- c(18000,22000,24000,26000,28000,30000,34000,42000,85000,200000)

# Histogram

hist(income,

   col="lightgreen",

   main="Histogram of Monthly Income",

   xlab="Monthly Income (Rs)")


grid()

# Q-Q Plot

qqnorm(income,

    main="Normal Q-Q Plot of Monthly Income")

qqline(income,

    col="red",

    lwd=2)

grid()
```

Interpretation

Histogram

- Most incomes fall between **₹18,000 and ₹42,000**.
- Two very high incomes (**₹85,000 and ₹200,000**) create a long right tail.
- Therefore, the histogram is **highly positively skewed**.

Q-Q Plot

- Points deviate considerably from the straight reference line.
- The upper-right points lie far above the line due to the very high incomes.
- This indicates that the data is **not normally distributed**.

Conclusion

The monthly income data is **highly positively skewed** rather than normally distributed. The histogram shows a long right tail, and the Q-Q plot confirms this by displaying significant departures from the reference line.